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Advanced Computer Architecture

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Name
Last Name

The student has to upload 1 PDF files using the Microsoft Form shared via the Zoom chat at the beginning of the exam. Few notes about the file:

1. You have to submit 1 PDF
2. The PDF has to be max 10MB
3. The PDF has to be named WITH ONLY your codice persona, e.g., 10000000.pdf (the system will automatically add your name to the file name)
4. The PDF has to report your name and the codice persone on the first page
5. All these points are in logic AND, if one of them will not be respected, the ENTIRE exam will not be evaluated

Problem

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	5	6	7	8
MULTD F0 F4 F2	6	9	19	20
ADDD F2 F8 F6	7	9	10	11
DIVD F12 F0 F6	8	21	31	32
SUBD F6 F8 F2	9	12	13	14

Question A. Is there a “configuration” that can respect the shown execution?
If the previous answer was positive, how many units? Which kind? What latency?

Question B. If the previous table was not correct, please, write the right one
and specify the number, kind and latency for each unit.

Answer. A

Answer. B

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2				
LD F2 45+ R3				
MULTD F0 F4 F2				
ADDsD F2 F8 F6				
DIVD F12 F0 F6				
SUBD F6 F8 F2				